THE $e_k \star e_r$ PIERI RULE FOR ALL k: PARABOLIC KERNEL, CHAIN FACTORIZATION, AND A ONE-INDEX q-SERIES IDENTITY

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ABSTRACT. For every $k \geq 1$, $r \geq 0$ and every number of variables $m \geq 0$, we prove

$$t^{-\binom{k}{2}} e_k(Y) \bullet e_r = \sum_{b=0}^{k} s^b \, F_{k-b}(t^{r-b}) \, e_b \, e_{r+k-b}, \qquad s = q^{-1},$$

with the explicit polynomial F_n defined in (1) below. Via Hikita's Lemma 3.1, Definition 3.4 and Lemma 3.3 (arXiv:2503.23597), the left side is $e_k \star e_r$. The cases $k = 1$ and $k = 2$ recover Hikita's Theorem 3.12 and our W_r. The cases $k = 3$ and $k = 4$ recover the Day 193 and Day 195 closed forms, which are now proved for all r.

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(`proofs/2026-09-26-day207b-ek-star-er-general-k-PROVED.md`, `scripts/day207b/`)
Revised: 2026-09-30: Clio review (UID 301): ordering in $e_k(Y)$; $t = 0$ threshold $r \geq k - 1$.
2026-10-08: Clio review of 2026-10-05 (UID 325/326), F1–F4 below.
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Revision 2026-10-16. Clio's review of 2026-10-15 endorses Theorem 1 and all of §5 as proved (independent affine-Hecke reimplementation, 60/60, $m \leq 4$; three planted lesions detected; Lemma 7 checked step by step by hand). Changes:

G1. R0 withdrawn. Bijectivity of $q_{(m)}$ is Hikita's Lemma 3.1, so the $\star$-reading is unconditional. "Modulo R0" is gone from the abstract.
G2. The residue lemma is Macdonald III (2.9), not (2.10). (2.10) is the generating function $Q(u) = H(u)/H(tu)$.
G3. The $m = 0$ case of the Step Lemma now gets an actual argument. The old sentence cited q_n, which does not occur in T_k. The content of that case is $F_k(t^{-k}) = 0$.
G4. Added the Concha–Lapointe/Hikita Y_i dictionary (§6). Lemma $\star$ has been taken out of the machine-check list, since it is proved. The "Open" paragraph is now a Corollary.
G5. In the proof of the Step Lemma, it is now explicit which bracket Lemma $\star$ evaluates.

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Revision 2026-10-08. Clio (review of 2026-10-05) found four losses between the proof source (`proofs/2026-09-26-day207b-ek-star-er-general-k-PROVED.md`, ll. 190–210) and the 2026-09-30 PDF, all in the proof of the Step Lemma (§5). The mathematics was correct in the source; each item below is restored from it. Thanks to Clio.

F1. "apply Lemma 2" pointed at this note's Key Lemma. Now reads *Day 205b Lemma 2* (the residue lemma $(1-t)\sum_i X_i^n \prod_{j\neq i} a_{ij} = q_n$, $n \geq 1$), stated in place.

F2. The hypothesis $g_c(0) = 0$ was dropped; without it the Ω-lemma is false (off by $[m]_t$ at $n = 0$). Restored, with the check that both pieces of the split vanish at $u = 0$.

F3. Restored "coefficientwise in z" (justifying Ω on g_c, which has a pole at $u = -1/\gamma$) and "for $m = 0$ both sides are 0".

F4. In (b), the manifestly polynomial form $[w^b]P = \sum_{n>b}(\dots)$ is now printed first, with the reason it equals $-\sum_{n\leq b}(\dots)$.

Also restored the label (R) on the Recursion lemma, and the scope of the $\star$-reading (Clio, UID 326): the operator identity of Theorem 1 is proved; its reading as $e_k \star e_r$ rests on Hikita Def. 3.4/Lemma 3.3 *and* bijectivity of $q_{(m)}$ (R0), which is not proved here. *[Withdrawn 2026-10-16: R0 is Hikita's Lemma 3.1; see G1.]*

1. Conventions and statement

Operators. The conventions are those of the W_r note.

$$T_i F = t\, s_i F + (t-1) X_{i+1} \frac{s_i F - F}{X_i - X_{i+1}}, \qquad \pi F = X_1 F(X_2, \dots, X_m, sX_1),$$

$$Y_i = t^{m-i} T_{i-1} \cdots T_1\, \pi\, T_{m-1}^{-1} \cdots T_i^{-1}, \qquad e_k(Y) := \sum_{1 \leq b_1 < \cdots < b_k \leq m} Y_{b_1} Y_{b_2} \cdots Y_{b_k},$$

where each product is taken in increasing index order from left to right, so Y_{b_k} acts first. The proof of Theorem 1 uses only this order and never uses commutativity of the Y's, so the ordering is part of the definition.

Symmetric functions.

- $a_{ij} = (X_i - tX_j)/(X_i - X_j)$ and $\prod_A^{\times} = \prod_{i\in A,\, j\notin A} a_{ij}$.
- $E(z) = \prod_i (1 + X_i z) = \sum_n e_n z^n$.
- $Q(y) = \sum_n q_n y^n = E(-ty)/E(-y)$.
- $[n] = [n]_t$.

Coefficients. For $0 \leq j \leq n$ put

(1)

$$\alpha_j = \frac{\prod_{i=1}^{j}(s - t^i)}{(t;t)_j}, \quad \alpha_{-1} = 0, \qquad c(n,j) = \frac{(s;t)_{n-j}}{(t;t)_{n-j}} \big(\alpha_j - s\, t^{n-j} \alpha_{j-1}\big), \qquad F_n(w) = \sum_{j=0}^{n} c(n,j)\, w^j.$$

Set $c(n,j) = 0$ outside the range $0 \leq j \leq n$.

Theorem 1. *For all $m \geq 0$, $k \geq 1$ and $r \geq 0$,*

$$\sum_{r\geq 0} z^r\, t^{-\binom{k}{2}} e_k(Y) \bullet e_r = \sum_{j+b\leq k} s^b t^{-kj} c(k-b, j)\; e_b\, z^{b-k} E(t^j z).$$

Taking the coefficient of z^r gives

$$t^{-\binom{k}{2}} e_k(Y) \bullet e_r = \sum_{b=0}^{k} s^b F_{k-b}(t^{r-b})\, e_b e_{r+k-b}.$$

2. Hecke facts

The T_i satisfy the braid relations and $(T_i - t)(T_i + 1) = 0$. Hence:
- T_w is well defined on reduced words;
- $T_u T_v = T_{uv}$ whenever $\ell(uv) = \ell(u) + \ell(v)$.

For $J \subseteq S$, let W^J be the set of minimal left coset representatives, so that $w = w^J w_J$ uniquely and length-additively.
(Chain). If $K \subseteq J$, then $W^J \times (W_J)^K \to W^K$, $(u, v) \mapsto uv$, is a length-additive bijection.
- If $s \in K$, then $\ell(uvs) = \ell(u) + \ell(vs) = \ell(u) + \ell(v) + 1$.
- Conversely, if $x \in W^K$ but $x_J s < x_J$ for some $s \in K$, then $\ell(xs) < \ell(x)$, which is a contradiction.

Spherical vectors. If $T_k H = tH$ for all $s_k \in J$, then $T_v H = t^{\ell(v)} H$ for every $v \in W_J$.
Shifting through π. $\pi T_k = T_{k+1}\pi$ for $1 \le k \le m-2$.
Minimal representatives. Let $J_n = \langle s_i : i \neq n \rangle$. For $D = \{d_1 < \cdots < d_n\}$, the word

$$w(D) = (s_{d_1-1} \cdots s_1)(s_{d_2-1} \cdots s_2) \cdots (s_{d_n-1} \cdots s_n)$$

sends $l \mapsto d_l$. Its length is $\sum_l (d_l - l) = \ell(w^D)$. Hence it is a reduced word for the minimal representative w^D. We set $\sigma^{(k)} = \sum_{|D|=k} T_{w(D)}$.

3. The parabolic kernel (A_k)

Lemma 2 (Key Lemma)**.** *Let $1 \le c < d_1 < \cdots < d_n \le m$, and let H satisfy $T_k H = tH$ for all $k \neq n$. Then*

$$(T_c \cdots T_{m-1})^{-1} T_{w(D)} H = t^{\,n-(m-c)}\, T_{w(D-1)} H.$$

Proof. Let $x = s_c \cdots s_{m-1}$. It sends $m \mapsto c$ and $i \mapsto i+1$ for $c \le i < m$. Let $y = w(D-1)$; then $y(m) = m$.
- *Length.* x is increasing on $[1, m-1]$, and $xy(m) = c$ is inverted with exactly the $m - c$ positions i with $y(i) \ge c$. So $\ell(xy) = \ell(y) + m - c$.
- *Coset.* $xy([n]) = x(D-1) = D$. Therefore $xy = w^D v$ with $v \in W_{J_n}$ and $\ell(v) = m - c - n$.

Hence $T_x T_y H = T_{w^D} T_v H = t^{m-c-n} T_{w^D} H$. □

Proposition 3 (A_k)**.** *Let F be symmetric and $b_1 < \cdots < b_k$. Then $Y_{b_1} \cdots Y_{b_k} F = t^{\binom{k}{2}} T_{w(b)} \pi^k F$. Consequently $e_k(Y) F = t^{\binom{k}{2}} \sigma^{(k)} \pi^k F$.*

Proof. Induct on k. Put $H = \pi^n F$; it is symmetric in the head and in the tail. By the Key Lemma,

$$Y_c\, T_{w(D)} H = t^{m-c} T_{c-1} \cdots T_1\, \pi\, (T_c \cdots T_{m-1})^{-1} T_{w(D)} H = t^n\, T_{c-1} \cdots T_1\, \pi\, T_{w(D-1)} H.$$

Shifting through π turns this into $t^n T_{w(\{c\} \cup D)} \pi^{n+1} F$. Finally, $\binom{n}{2} + n = \binom{n+1}{2}$. □

4. The k-set kernel (K_k)

Let $\sigma_{[l,m]} = \sum_{a=l}^{m} T_{a-1} \cdots T_l$ and $K_l = \langle s_{l+1}, \ldots, s_{m-1} \rangle$.
(C1) Two factorizations. Applying (Chain) along $K_k \subset K_{k-1} \subset \cdots \subset K_0$, and also along $K_k \subset J_k$ (where $(W_{J_k})^{K_k} = S_k$), gives

$$\sum_{u \in W^{K_k}} T_u = \sigma_{[1,m]} \cdots \sigma_{[k,m]} = \sigma^{(k)} \sum_{v \in S_k} T_v.$$

Consequently, if G is symmetric in the head and in the tail, then $\sigma_{[1,m]} \cdots \sigma_{[k,m]} G = [k]!\, \sigma^{(k)} G$.
(C2) Ordered formula. Suppose G is symmetric in the tail only. Iterating Day 205b Lemma 1, applied first in $X_2, \ldots, X_m$ over $\mathbb{Q}(s,t)(X_1)$ and then at the top level, gives

$$\sigma_{[1,m]} \cdots \sigma_{[k,m]} G = \sum_{(a_1,\ldots,a_k) \text{ distinct}} G(X_{a_1}, \ldots, X_{a_k}; \text{rest}) \prod_{l=1}^{k} \prod_{j \notin \{a_1,\ldots,a_l\}} a_{a_l j}.$$

(C3) Poincaré identity. Taking $m = k$ and $G = 1$ in (C1) and (C2) gives $\sum_{w\in S_k}\prod_{l<l'} a_{w_l w_{l'}} = [k]!$.

Proposition 4 (K_k)**.** *If G is symmetric in the head and in the tail, then $\sigma^{(k)}G = \sum_{|A|=k} G^{(A)}\prod_A^{\times}$.*

Proof. In (C2), group the ordered tuples by their underlying set A. The weight of each tuple factors as $\prod_A^{\times}\cdot\prod_{l<l'} a_{a_l a_{l'}}$. Summing over the orderings of A gives $[k]!\prod_A^{\times}$ by (C3). Now compare with (C1). □

5. The generating function (E_k)

Let $\varphi(u) = u(1+suz)/(1+uz)$. By ($A_k$) and ($K_k$),

$$\Gamma_k := \sum_r z^r t^{-\binom{k}{2}} e_k(Y)\bullet e_r = E(z)\sum_{|A|=k}\prod_A^{\times}\prod_{a\in A}\varphi(X_a).$$

This holds for all $m \geq 0$; both sides vanish when $m < k$. Let T_k denote the right side of Theorem 1.

Lemma 5 (Recursion (R))**.** *For $k, m \geq 1$,*

$$[k]\,\Gamma_k(X) = \sum_i X_i(1+sX_iz)\prod_{j\neq i} a_{ij}\,\Gamma_{k-1}(\widehat{X}_i).$$

Proof. Use (C2) and (C3), and peel off the first index $a_1 = i$. Then use $\varphi(X_i)E(X;z) = X_i(1+sX_iz)E(\widehat{X}_i;z)$. □

Lemma 6 (Step)**.** *For every $m \geq 0$ and $k \geq 1$,*

$$\sum_i X_i(1+sX_iz)\prod_{j\neq i} a_{ij}\,T_{k-1}(\widehat{X}_i) = [k]\,T_k(X).$$

Proof. Let $\gamma = t^j z$. Expand

$$e_b(\widehat{X}_i) = \sum_c (-X_i)^c e_{b-c}, \qquad E(\widehat{X}_i;\gamma) = \frac{E(\gamma)}{1+\gamma X_i},$$

Then the term $N'_{jb}e_b(\widehat{X}_i)z^{b-k+1}E(\widehat{X}_i;\gamma)$ of $T_{k-1}(\widehat{X}_i)$, multiplied by $X_i(1+sX_iz)$, becomes

$$N'_{jb}\,z^{b-k+1}E(\gamma)\sum_c(-1)^c e_{b-c}\,g_c(X_i), \qquad g_c(u) = \frac{u^{1+c}(1+suz)}{1+\gamma u},$$

and we split

$$g_c(u) = st^{-j}u^{1+c} + (1-st^{-j})\frac{u^{1+c}}{1+\gamma u}, \qquad \text{using} \qquad \frac{1+suz}{1+\gamma u} = st^{-j} + \frac{1-st^{-j}}{1+\gamma u}.$$

The coefficients e_{b-c} are now fully symmetric, and $g_c(0) = 0$; indeed both pieces of the split vanish at $u = 0$, since each carries the factor u^{1+c} with $c \geq 0$. So *Day 205b Lemma 2* (the residue lemma $(1-t)\sum_i X_i^n\prod_{j\neq i} a_{ij} = q_n$ for $n \geq 1$, i.e. Macdonald III (2.9); not Lemma 2 of this note) applies *coefficientwise in z*:

$$\sum_i g(X_i)\prod_{j\neq i} a_{ij} = (1-t)^{-1}\Omega[g], \qquad \Omega[u^n] := q_n \quad (n\geq 1), \qquad \text{valid for } g(0) = 0.$$

Here "coefficientwise in z" is what licenses applying Ω to g_c, which has a pole at $u = -1/\gamma$: since $\gamma = t^j z$, each coefficient of z^N in $g_c(u)$ is a polynomial in u with zero constant term. The hypothesis $g(0) = 0$ is load-bearing: for $g = 1$ one gets $(1-t)\sum_i\prod_{j\neq i} a_{ij} = 1 - t^m$, not $q_0 = 1$ (off by $[m]_t$). Two facts evaluate the result:

(a) *The s-part.* $\sum_c(-1)^c e_{b-c}q_{c+1} = (1-t^{b+1})e_{b+1}$; this is $[y^{b+1}]$ of $Q(y)E(-y) = E(-ty)$.

(b) *The pole part.* Since $\sum_c (-1)^c e_{b-c} u^c = [w^b]E(w)/(1+uw)$, and $\Omega[u/((1+wu)(1+\gamma u))] = \sum_{a,p\geq 0}(-w)^a(-\gamma)^p q_{1+a+p} = (Q(-\gamma)-Q(-w))/(w-\gamma)$, using $E(x)Q(-x) = E(tx)$ twice gives

$$E(\gamma)\sum_c (-1)^c e_{b-c}\,\Omega\left[\frac{u^{1+c}}{1+\gamma u}\right] = [w^b]\,P(w,\gamma), \qquad P(w,\gamma) := \frac{E(w)E(t\gamma)-E(\gamma)E(tw)}{w-\gamma}.$$

P is a polynomial, since its numerator vanishes at $w=\gamma$. Writing the numerator as $\sum_n e_n w^n\big(E(t\gamma)-t^nE(\gamma)\big)$ gives the manifestly polynomial form

$$[w^b]P = \sum_{n>b} e_n \gamma^{n-1-b}\big(E(t\gamma)-t^nE(\gamma)\big) = -\sum_{n\leq b} e_n\gamma^{n-1-b}\big(E(t\gamma)-t^nE(\gamma)\big).$$

The two forms agree because the full sum over n is $\gamma^{-1-b}\big(E(\gamma)E(t\gamma)-E(t\gamma)E(\gamma)\big)=0$. The second form displays only apparent poles γ^{n-1-b}; the first shows there are none.

Write $N'_{jb} = s^b t^{-(k-1)j} c(k-1-b,j)$ for the coefficients of T_{k-1}, and put $n = k-b'$. The coefficient of $e_{b'}z^{b'-k}E(t^jz)$ on the left side is then $(1-t)^{-1}s^{b'}t^{-kj}$ times

$$(1-t^{b'})c(n,j) + t^{b'}\Big[(1-st^{-j})\sum_{p\geq 0}(st^{-j})^p c(n-1-p,j) - t^n(1-st^{1-j})\sum_{p\geq 0}(st^{1-j})^p c(n-1-p,j-1)\Big].$$

By Lemma 7 the square bracket equals $(1-t^n)c(n,j)$. So the whole expression is $(1-t^{b'})c(n,j) + t^{b'}(1-t^n)c(n,j) = (1-t^k)c(n,j)$, which is the claim.

The case $m=0$. This case is not trivial. Every equality above (the residue lemma, (a), (b), both forms of $[w^b]P$, and the final collection via Lemma 7) is an identity between expressions in the e_n and $E(\cdot)$ that holds in every Λ_m. Specialising to $m=0$, i.e. $e_n \mapsto 0$ for $n\geq 1$ and $E\mapsto 1$, is a ring homomorphism, so the identity specialises. The left side becomes the empty sum, and $[k]T_k$ becomes $[k]z^{-k}F_k(t^{-k})$. So the case $m=0$ of the lemma says exactly $F_k(t^{-k})=0$. (This argument is Clio's, review of 2026-10-15, §6 Finding 4. The sentence it replaces cited q_n, which does not occur in T_k.) □

Lemma 7 ($\star$)**.** *For $n\geq 1$ and $j\geq 0$,*

$$(1-t^n)c(n,j) = (1-st^{-j})\sum_{p\geq 0}(st^{-j})^p c(n-1-p,j) - t^n(1-st^{1-j})\sum_{p\geq 0}(st^{1-j})^p c(n-1-p,j-1).$$

Proof. Generating functions. Let $G(x) = \sum_n \frac{(s;t)_n}{(t;t)_n}x^n$. Comparing coefficients gives $(1-x)G(x) = (1-sx)G(tx)$. Let $C_j(x) = \sum_n c(n,j)x^n$. Then

$$C_j(x) = x^j\big[\alpha_j G(x) - s\alpha_{j-1}G(tx)\big].$$

Reformulation. In terms of the C_j, the claim is

$$\frac{C_j(x)(1-x)}{1-st^{-j}x} - C_j(tx) = -\frac{(1-st^{1-j})\,tx\,C_{j-1}(tx)}{1-st^{2-j}x}.$$

Linear factorization. Put $D_j = \alpha_j - s\alpha_{j-1}$. Since $\alpha_j = \alpha_{j-1}(s-t^j)/(1-t^j)$, we have

$$\alpha_j(1-sy) - s\alpha_{j-1}(1-y) = D_j(1-st^{-j}y).$$

Indeed, both sides are linear in y and agree at $y=0$ and at $y=t^j/s$.

Reduction. Substituting the factorization, both sides of the reformulated claim become the common factor $x^jG(t^2x)/(1-tx)$ times

$$D_j(1-t^j) \qquad \text{and} \qquad t(s-t^{j-1})D_{j-1}$$

respectively. These are equal, because $D_j = \alpha_{j-1}t^j(s-1)/(1-t^j)$ for $j\geq 1$. For $j=0$ both sides vanish. □

Proof of Theorem 1. Induct on k, for all m simultaneously. The base case is $\Gamma_0 = E(z) = T_0$. For $k \geq 1$:

- if $m = 0$, then $\Gamma_k = 0$, and the Step Lemma (its case $m = 0$, i.e. $F_k(t^{-k}) = 0$) gives $T_k = 0$;
- if $m \geq 1$, combine the Recursion, the induction hypothesis in $m-1$ variables, and the Step Lemma.

Finally, $[z^r]\, z^{b-k} E(t^j z) = t^{j(r+k-b)} e_{r+k-b}$. □

6. Remarks

What is classical and what is new. The kernel form of $e_k(Y)$ is classical in the q-shift DAHA; see Concha–Lapointe, arXiv:2307.02385, Lemmas 8 and 10. Their Y_i is the mirror image of Hikita's (and ours). Per Clio's reading of their §4 (review of 2026-10-15, §7), CL's operator is $Y_i^{\mathrm{CL}} = t^{-N+i}\, T_i \cdots T_{N-1}\, \omega\, \bar{T}_1 \cdots \bar{T}_{i-1}$, while ours is $Y_i = t^{m-i}\, T_{i-1} \cdots T_1\, \pi\, T_{m-1}^{-1} \cdots T_i^{-1}$. CL's string ascends where ours descends, and they have ω where we have π. Reversing the variables, $x_i \leftrightarrow x_{N+1-i}$ (so $T_j \leftrightarrow T_{N-j}$), takes our Y_{N+1-i} to an operator of CL's shape with prefactor t^{i-1}. So the dictionary is $i \leftrightarrow N+1-i$, up to a global power of t and the normalisations of ω and $\bar{T}_j$, which we have not re-checked first-hand against CL. On symmetric functions, $e_r(Y)$ is invariant under reindexing, so Lemma 10 transfers once the scalar is matched. This dictionary is a different one from the DFK relation $Y_i = \theta^{m-1}(Y_{i,-1})^{-1}$ in the long version. The level-one π^k transfer and the e-basis Pieri rule for general k are, as far as the Day 207 audit found, new.

Specializations. The following are checked symbolically in $u = t^r$: $k = 1$ gives Hikita's Theorem 3.12; $k = 2$ gives W_r; $k = 3, 4$ give Days 193 and 195. The case $r = 0$ recovers Hikita's Lemma 3.3, $e_k(Y) \bullet 1 = t^{\binom{k}{2}} e_k$; this is checked symbolically for $k \leq 7$.

Machine checks. (`scripts/day207b/`, all OK). Lemma $\star$ is not on this list because it is proved in §5; the symbolic run for $n \leq 9$ is consistency evidence only. Clio extended it to $n \leq 30$ at eight exact points.

- Key Lemma and (A_k): exact for $m \leq 6$.
- (C2): checked.
- Step Lemma: checked with z symbolic.
- Full theorem against a direct AHA computation: 216 cases, including $m < k$.

The $t = 0$ specialization (erratum). The Corollary, proved in `proofs/2026-09-26-day207b-ek-star-er-general-` says that at $t = 0$

$$e_k \star e_r = \sum_{b<\mu} (1-s) s^b e_b e_{r+k-b} + s^\mu e_\mu e_M, \qquad \mu = \min(k,r),\ M = \max(k,r).$$

The uncollapsed form $\sum_{b<k}(1-s)s^b e_b e_{r+k-b} + s^k e_k e_r$ holds for $r \geq k-1$. The P.S. of the covering email (UID 732) said $r \geq k$, which gave away one case. At $r = k-1$, the $b = k-1$ and $b = k$ terms combine to $s^{k-1} e_{k-1} e_k$. At $r = k-2$, the uncollapsed form is off by $(1-s)s^{k-1}(e_{k-1}^2 - e_{k-2}e_k) \neq 0$. Clio checked this for $k \leq 6$ (review of 2026-09-29, UID 301), and we re-checked it for $k \leq 5$.

Corollary (forced roots; suggested by Clio). For $k \geq 1$ and $1 \leq N \leq k$,

$$\sum_{b=0}^{k-N} s^b\, F_{k-b}(t^{-N-b})\, e_b\, e_{k-N-b} = 0 \quad \text{in } \Lambda; \qquad \text{in particular } (N=k) \quad F_n(t^{-n}) = 0 \ \ (n \geq 1).$$

Proof. Γ_k is a power series in z, so $[z^{-N}]T_k = 0$ by Theorem 1. Since $[z^{-N}]z^{b-k}E(t^j z) = t^{j(k-N-b)} e_{k-N-b}$, summing over j gives the display. □

So $1 - t^n w$ divides $F_n(w)$ for every n. This is proved, not merely computed. It is the case $m = 0$ of the theorem. We checked the full family for $k \leq 8$ and every N, with the e's independent (`scripts/day242/clio_corollary_check.py`, 121/121, lesion controls firing). The quotient's ${}_2\phi_1$ form, $F_n(w)/(1-t^n w) = \frac{(s;t)_n}{(t;t)_n}\, {}_2\phi_1(t^{1-n}, t/s; t^{1-n}/s; t; w)$, is still only computed: $n \leq 6$ here, $n \leq 14$ by Clio. It is not needed for the theorem.